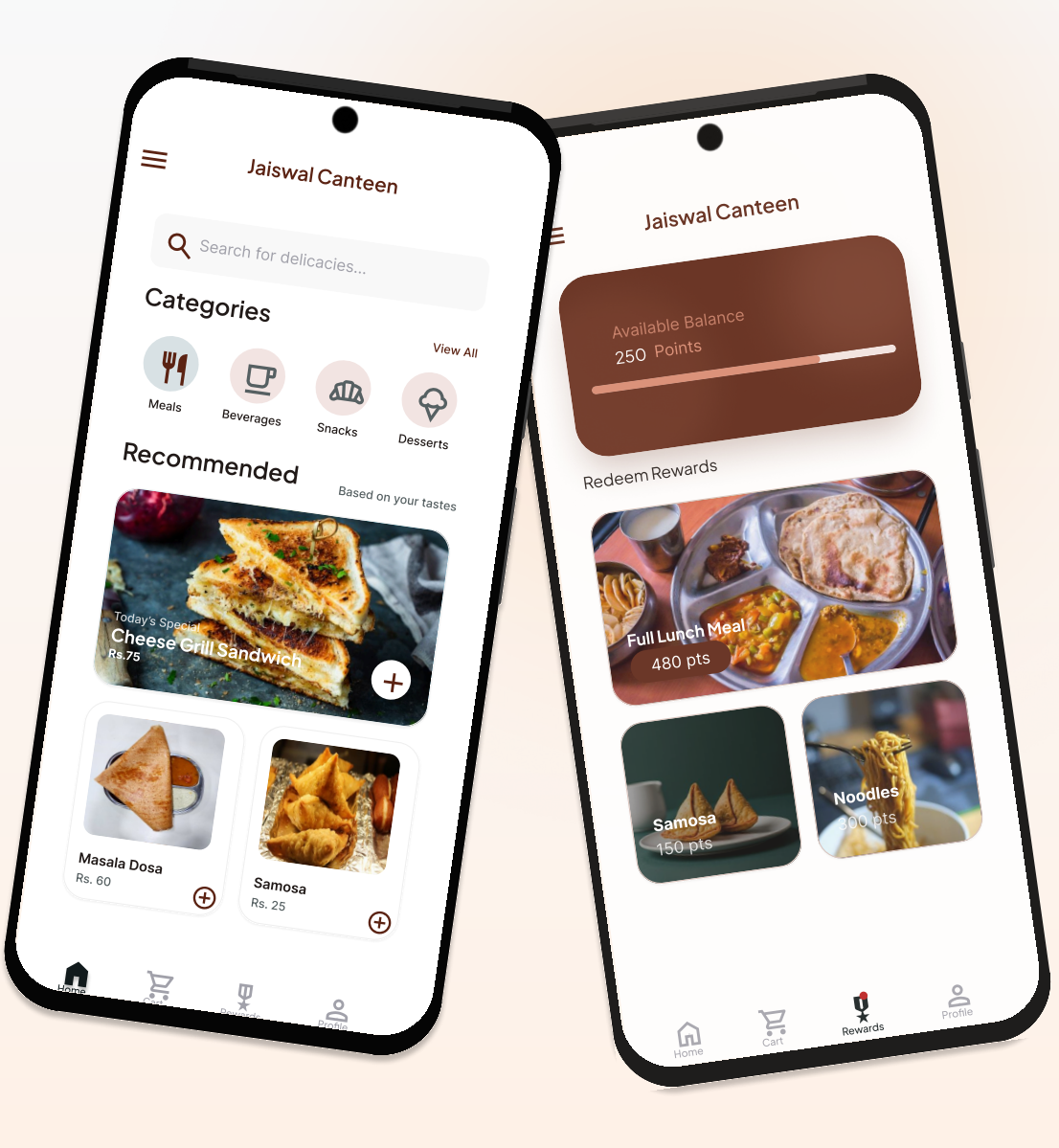
UX CASE STUDY

Smart Canteen Experience

Designing a seamless canteen experience to reduce queues, payment delays, and pickup confusion at Indus University.

Duration:	Role:	Tools:	Process:
4 Weeks	UX Designer	Figma	Design Thinking


Jaiswal Canteen
Search for delicacies...
Categories
View All
Meals
Beverages
Snacks
Desserts
Recommended
Based on your tastes
Today's Special
Cheese Grill Sandwich
Rs.75
Masala Dosa
Rs. 60
Samosa
Rs. 25
Jaiswal Canteen
Available Balance
250 Points
Redeem Rewards
Full Lunch Meal
480 pts
Samosa
150 pts
Noodles

Designed By:
Vrushti Lakhupota & Prachi Joshi

PROJECT OVERVIEW

Creating a Faster & Smarter Food Ordering System

The existing canteen experience at Indus University often leads to long queues, payment delays, and crowded pickup counters during peak hours. This project explores a smart digital solution to simplify ordering and improve student flow.

Problem

Students face long waiting times, crowded queues, and confusion during food ordering and payment at peak hours.

Goal

To design a queue-free digital canteen experience with faster ordering, seamless payment, and efficient pickup management.

Solution

A smart canteen system featuring Online ordering, live queue tracking, digital tokens, and cashless payments.

500+	12 Pm
Daily Students	Daily rush hour
10+ min	**UPI**
Average waiting time	Preferred payment

DESIGN PROCESS

Building a Smarter Canteen Experience Through UX

We followed a structured UX process to identify user pain points, explore ideas, and design a seamless canteen experience focused on reducing queues and payment friction

Research	Define	Ideate	Prototype	Test
Observed student behavior and identified major queue and payment issues.	Analyzed findings and framed the core usability problems.	Generated smart canteen concepts and digital flow solutions.	Created wireframes and interface concepts for the app system.	Evaluated usability and refined the experience based on feedback.

USER RESEARCH

Understanding Student Behavior & Canteen Challenges

To understand the existing canteen experience, we observed student behavior, conducted interviews, and analysed common issues faced during peak hours.

Observation

Observed student movement, ordering patterns, and crowd buildup during lunch hours.

Interviews

Spoke with students & manager to understand frustrations related to queues and payments.

Surveys

Collected feedback regarding waiting time, payment methods, and food pickup experience.

Key Research Insights

70%	10+ min
Students experience long queues during peak hours.	Average waiting time for ordering and payment.
UPI	**Crowded**
Most students prefer cashless payment methods.	Pickup area becomes congested after order completion.

Peak Hours
Payment Issues
Queue Delay
Student Flow

USER PERSONA

Meet Our Primary User

A quick look at the student users affected by queue and payment issues during canteen rush hours.

Kavya Sharma
- Age-20
- Student
- BBA

About

Visits the canteen daily during lunch breaks and prefers quick ordering with minimal waiting time

- Busy Schedule
- Tech Savvy
- Uses UPI

Goals
- Faster ordering
- Easy payment
- Less waiting

Frustrations
- Long Queues
- Payment Delays
- Crowded Pickup area

Meet Our Canteen Manager

Manoj Singh
- Age-32
- Daily Customers- 500+

About

Handles food orders, payment flow, and crowd management during peak canteen hours

- Manages Staff
- Handles Rush Hours
- Daily Operations

Goals
- Faster service flow
- Manage crowd efficiently
- Reduce payment confusion

Frustrations
- Long queues
- Delayed payments
- Pickup counter congestion

USER JOURNEY

Student Journey During Peak Hours

Students experience the most frustration during queue, payment, and pickup stages.

Steps	Actions	Thoughts/Feelings	Painpoints	Touchpoints	Opportunities
Hunger Trigger	Feels hungry during Class/ Break, Checks time	I'm starving, Slight excitement for break, Looking forward to food Excited	Uncertainty about crowd, Unsure if favourite item is available	Friends, College schedule, Canteen menu memory	Real-time crowd status, Live menu availability, Push notification for specials
Decide to eat	Chooses canteen, Decides what to eat mentally	Should I try something new?, Hope it's not too crowded. Confused	No updated menu info, No price clarity before reaching	Friends' suggestions, Past experience	Digital menu preview, Popular item suggestions, Combo recommendations
Travel to canteen	Walks to canteen, Talks with friends	Excited but slightly worried about queue Stressed	Reach and see long queue, No prior estimate of wait time	Physical campus pathway, Canteen entrance	Queue length display outside, Pre-order while walking, Estimated wait time shown
Decides to Eat	Stands in line, Checks menu board, Waits turn	This line is too long, Impatient, Slight irritation Irritated	Time waste, People cutting line	Physical queue, Menu board, Other customers Vendor counter	Digital menu screen, Token system, Engagement games while waiting
Place Order	Speaks order verbally, Repeats items if vendor mishears	Did they understand correctly?, Mild confusion Confused	Miscommunication, Noise	Vendor, Counter, Menu display	Digital order confirmation screen, App-based selection, Clear order summary
Wait for Food	Stands near pickup area,Listens for order number,Checks phone	Bored, Impatient, Hungry + annoyed Annoyed	No clear order tracking, Crowding, No notification when ready	Pickup counter, Crowd, Vendor calling out orders	App notification when ready, Gamified waiting experience, Digital token display
Pay & Eat	Pays cash or UPI, Waits for confirmation, Finds seat, Eats	Did my payment is complete?, Satisfied when the order is received Anxious	Slow network, Confusion between payment & pickup, Limited seating	QR code, Cash counter, UPI app, Seating area	Pre-payment before pickup, Auto payment confirmation, Seating availability indicator, Digital receipt

Canteen Staff Journey During Peak Hours

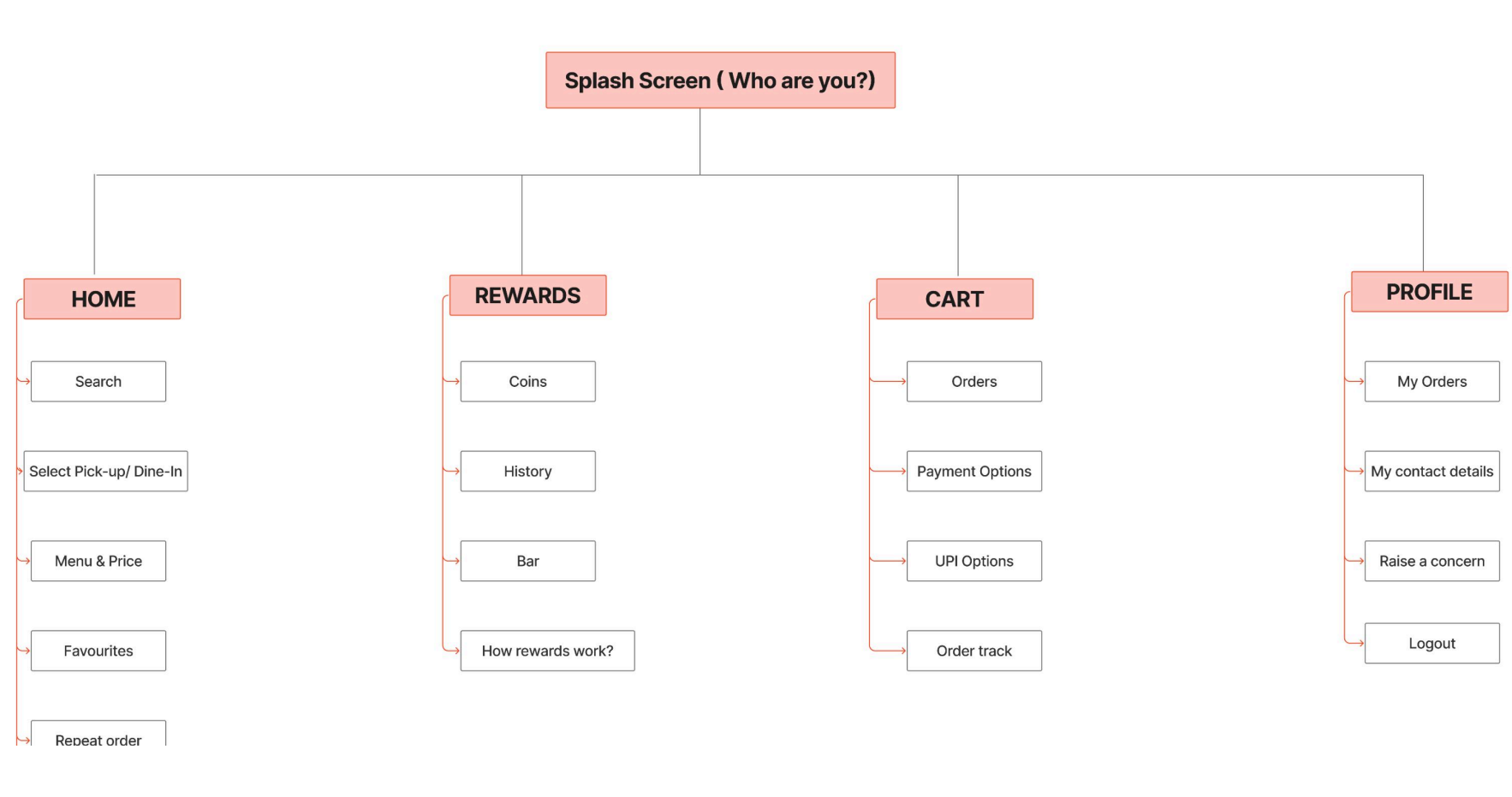
Staff struggle with managing orders, payments, and crowd flow during peak hours.

Steps	Actions	Thoughts/Feelings	Painpoints	Touchpoints	Opportunities
Morning Prep	Check Inventory, Prepare Ingredients	Do I have enough raw materials, How will we manage the rush today Stressed	End moment stock shortage, Too much rush	Kitchen staff / helpers, Canteen manager, Raw material suppliers	Check Inventory, Live inventory updates, Demand prediction, Preparation Insights
Peak Rush	Receive Orders, Accept Payments, Prepare and Dispatch Food	Orders are coming too fast, Hope there is no confusion and payment failures Anxious	Order Confusion, Delayed Payments, Overcrowding at canteen counter	Queue line, Menu board, Canteen manager	View and Track Orders, Auto status update, Auto payment confirmation
Serve Food	Handover Food, Manage crowd near counter	Did this person already pay?, Hope I didn't miss any order Stressed	Customers crowd near counter, Payment confirmation delays, Noise and chaos	Counter area, Queue line, Food Plates	Digital pickup screen display, Pickup confirmation, Order Status Board(Preparing........ Ready)
Payment	Announce total amount verbally, Accept cash/UPI, wait for customer to scan QR, Confirm transaction,	Did the payment actually go through?, Network is slow again..., I hope I don't miss a payment Anxious	Slow UPI, Cash handling errors,Sometimes there is no one at the cash counter	Personal smartphone (UPI verification), UPI apps (GPay, PhonePe, Paytm etc.), QR code stand, Calculator	Pre-paid ordering system, Real time payment confirmation
Inventory Check	Manual check of food items, Realize an item is almost finished, Quickly adjust preparation, Decides whether to discard	Which items are selling fast?, We didn't expect this much rush... Stressed	Manual stock tracking, Wastage unsold food	Excel sheet / notebook, Calculator, Daily sales record	Auto low-stock alert, End-of-day stock analytics, Wastage tracking report
CleanUp	Clear counter space, Dispose leftover food, Clean utensils & equipment, Wash trays / serving area	Today was exhausting, Did we sell properly? Tired & Anxious	No waste tracking system, No performance insight	Kitchen workspace, Storage shelves, Gas Stove, Sitting area, Kitchen Waste	Waste analysis dashboard, Cleaning checklist reminder, Suggested prep plan for next day
End of the Day	Count total cash manually, Cross-check UPI transactions on	Did the numbers match?, Did we make profit today?, Tomorrow we need to prepare more of that item.	Sometimes error in calculation, no data insights	Total sales record, Inventory report, Waste disposal	Auto-generated daily sales report, Profit & expense breakdown, Weekly performance dashboard

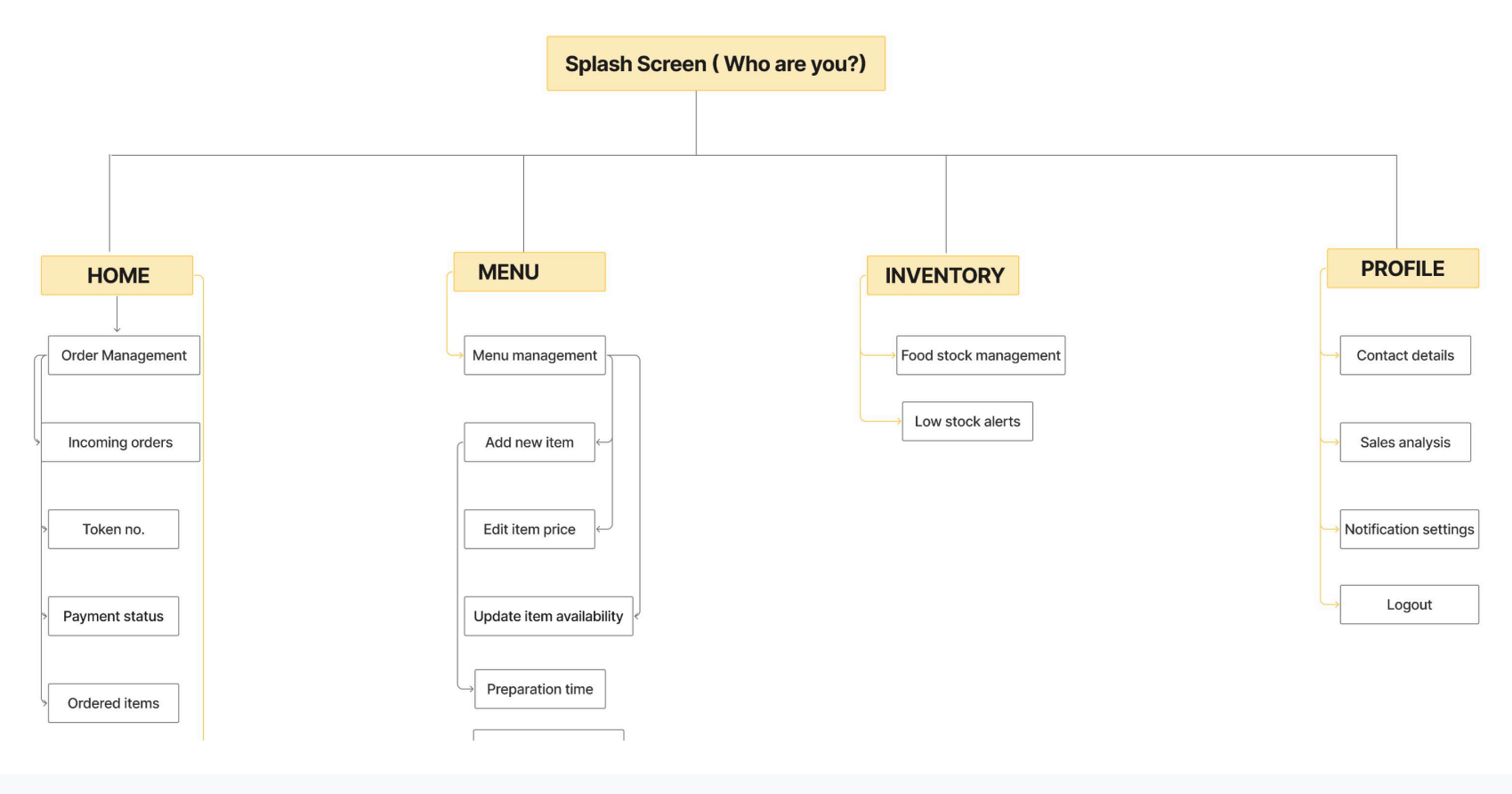
INFORMATION ARCHITECTURE

Smart Canteen Information Architecture for Customers

The information architecture organizes the ordering, payment, and pickup experience into a simple and seamless flow


Splash Screen (Who are you?)
HOME
Search
Select Pick-up/ Dine-In
Menu & Price
Favourites
Repeat order
REWARDS
Coins
History
Bar
How rewards work?
CART
Orders
Payment Options
UPI Options
Order track
PROFILE
My Orders
My contact details
Raise a concern
Logout

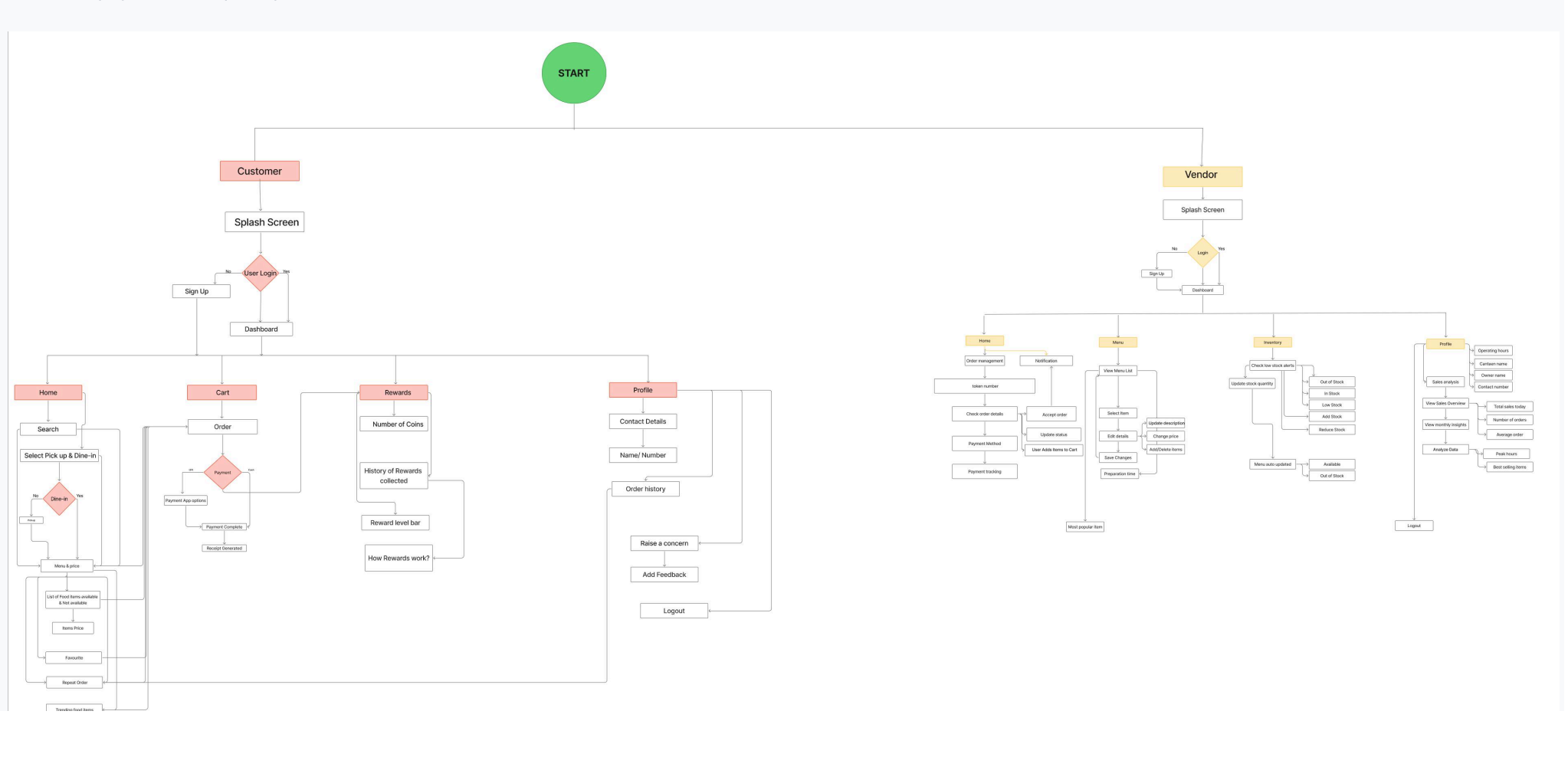
Smart Canteen Information Architecture for Canteen Staff


Splash Screen (Who are you?)
HOME
Order Management
Incoming orders
Token no.
Payment status
Ordered items
MENU
Menu management
Add new item
Edit item price
Update item availability
Preparation time
INVENTORY
Food stock management
Low stock alerts
PROFILE
Contact details
Sales analysis
Notification settings
Logout

USER FLOW

How users interact with the system step-by-step

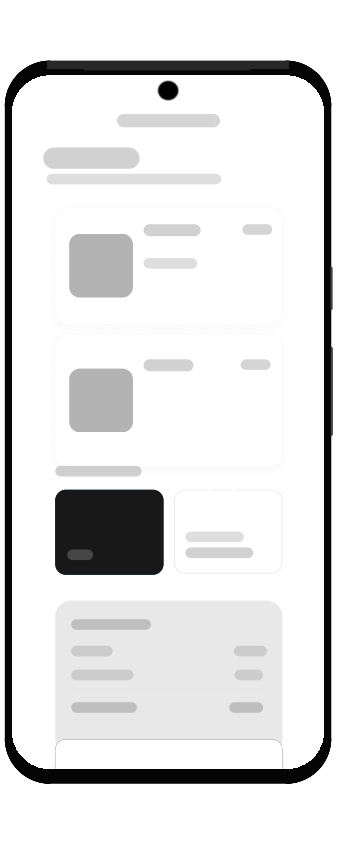
The user flow simplifies the canteen experience by reducing waiting time and streamlining payment and pickup.

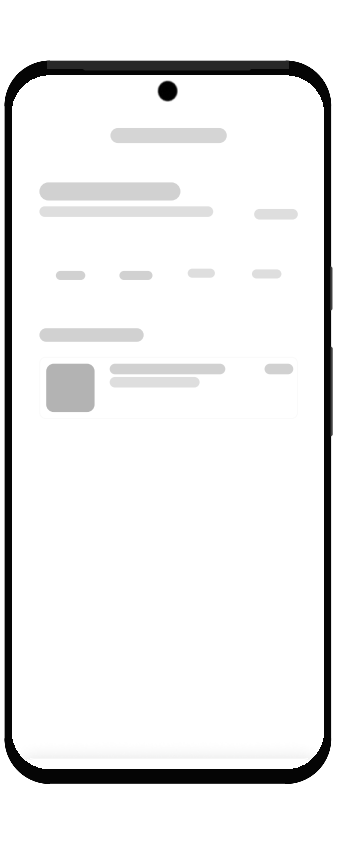

START
Customer
Splash Screen
User Login
Sign Up
Dashboard
Home
Search
Select Pick up & Dine-in
Cart
Order
Payment
Rewards
Number of Coins
History of Rewards collected
Reward level bar
How Rewards work?
Profile
Contact Details
Name/ Number
Order history
Raise a concern
Add Feedback
Logout
Vendor
Splash Screen

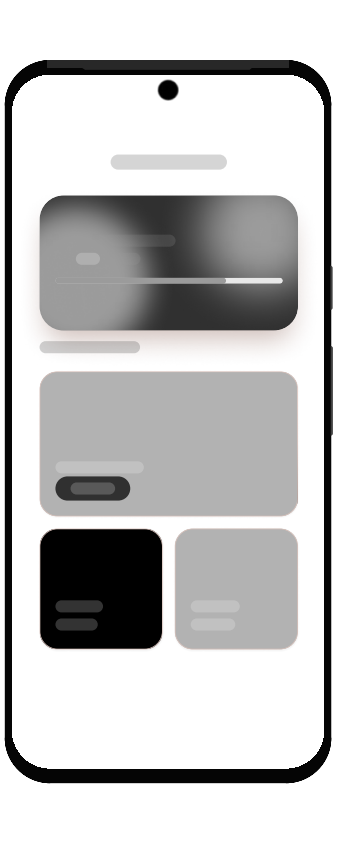
LOW FIDELITY WIREFRAMES

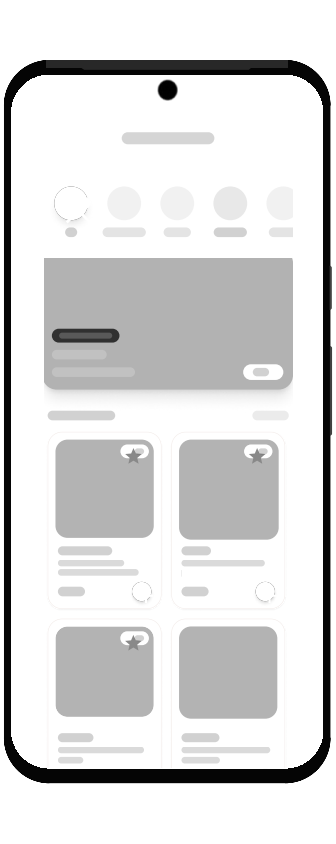
Exploring the Core User Experience

For Customers

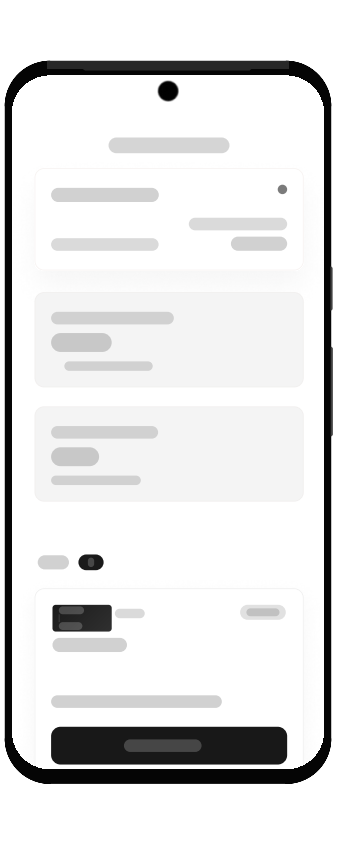


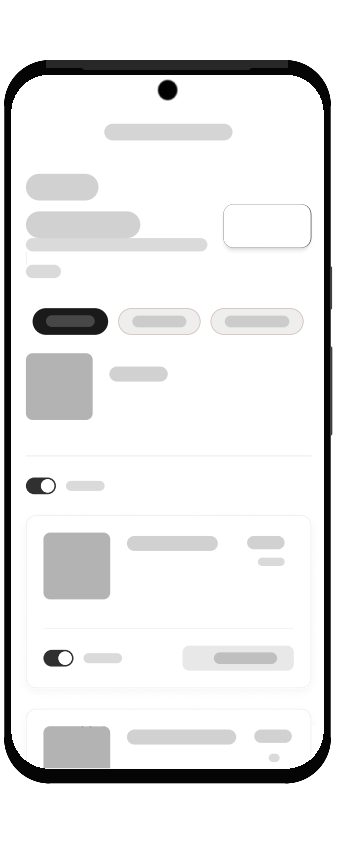


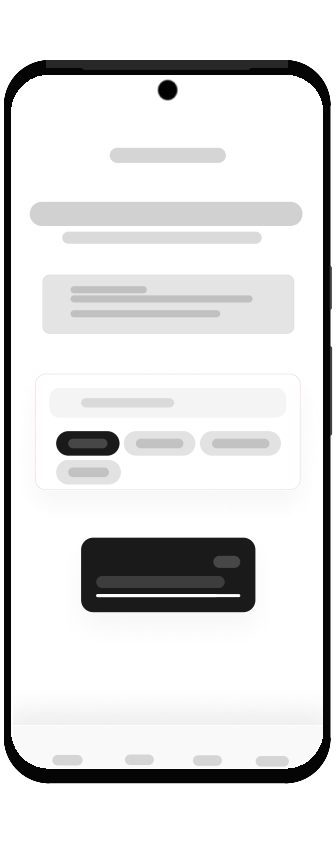




For Canteen Staff

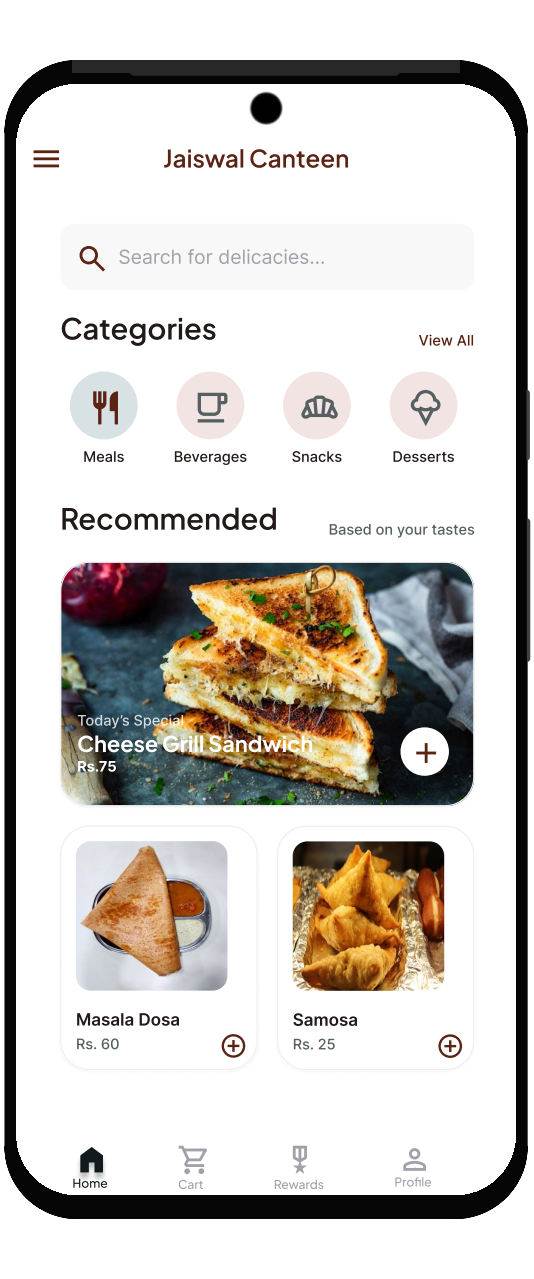


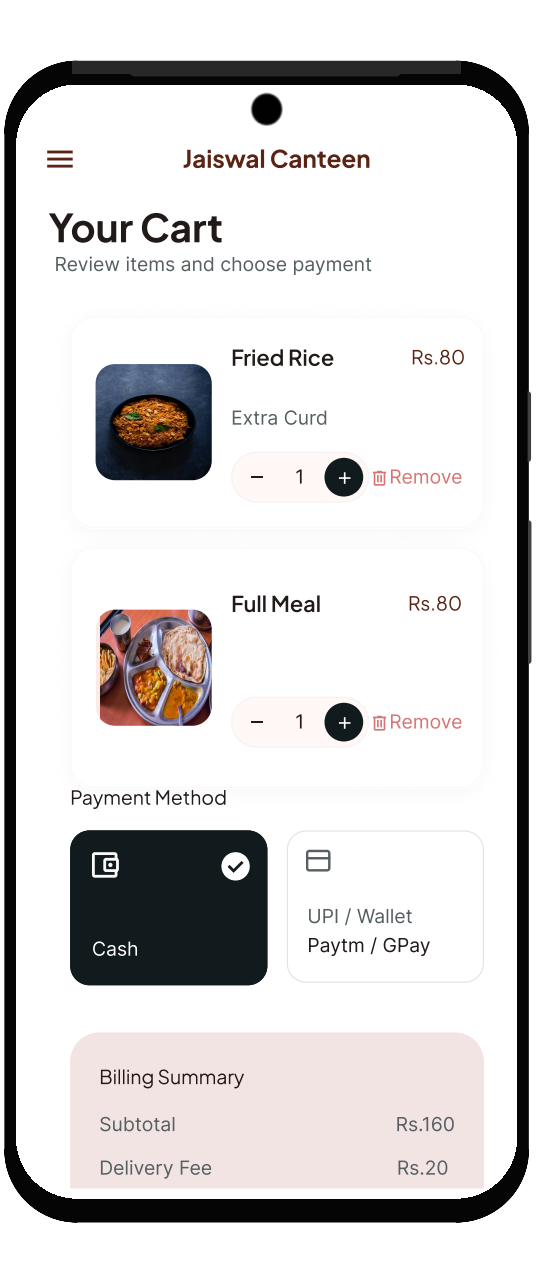


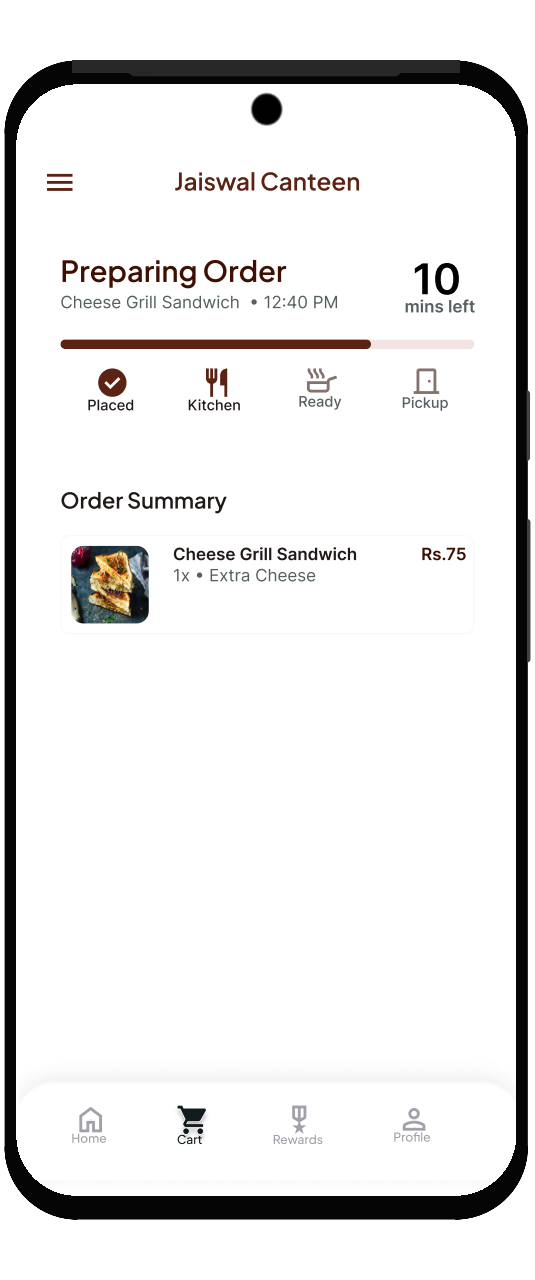


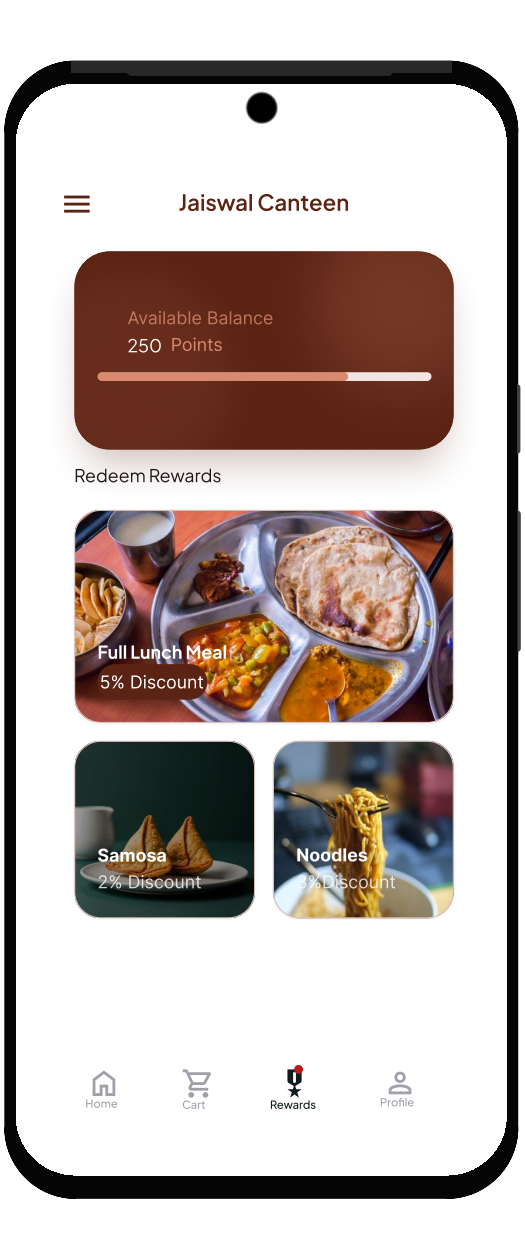
HI FIDELITY WIREFRAMES

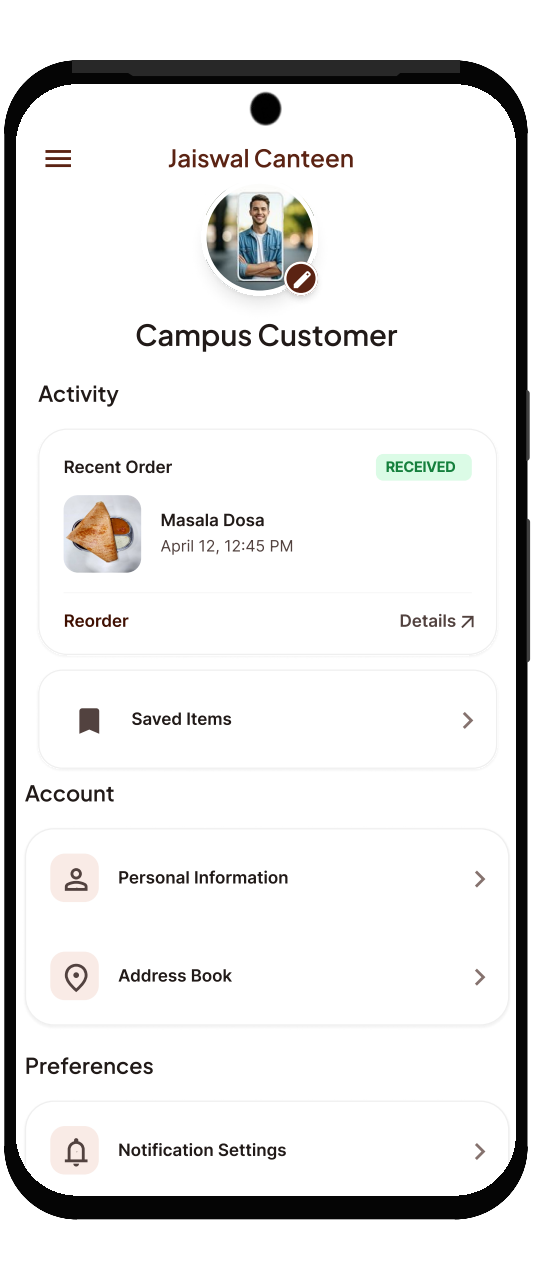
Designing a Seamless Smart Canteen Experience

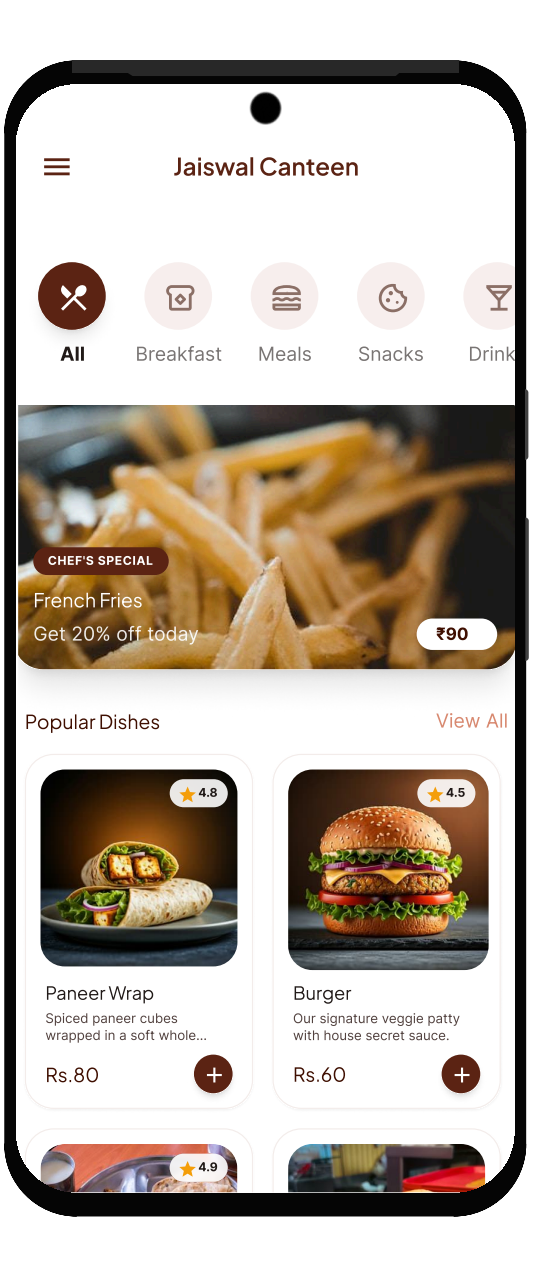

Jaiswal Canteen
Search for delicacies...
Categories
View All
Meals
Beverages
Snacks
Desserts
Recommended
Based on your tastes
Today's Special
Cheese Grill Sandwich
Rs.75
Masala Dosa
Rs. 60
Samosa
Rs. 25
Home
Cart
Rewards
Profile

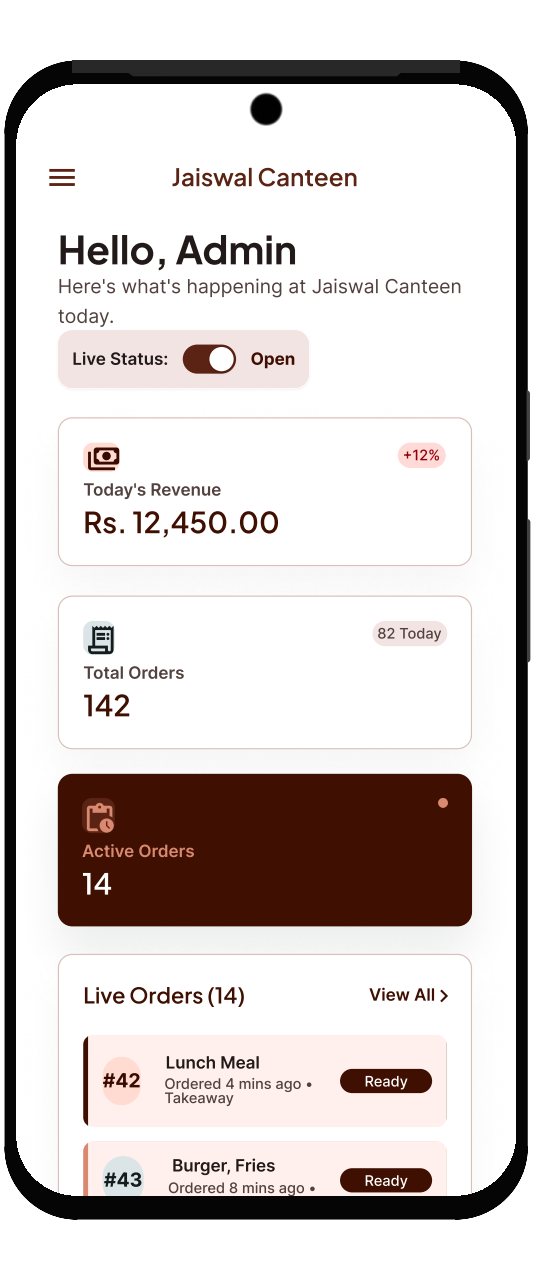

Jaiswal Canteen
Your Cart
Review items and choose payment
Fried Rice
Rs.80
Extra Curd
1
Remove
Full Meal
Rs.80
1
Remove
Payment Method
Cash
UPI / Wallet
Paytm / GPay
Billing Summary
Subtotal
Rs.160
Delivery Fee
Rs.20

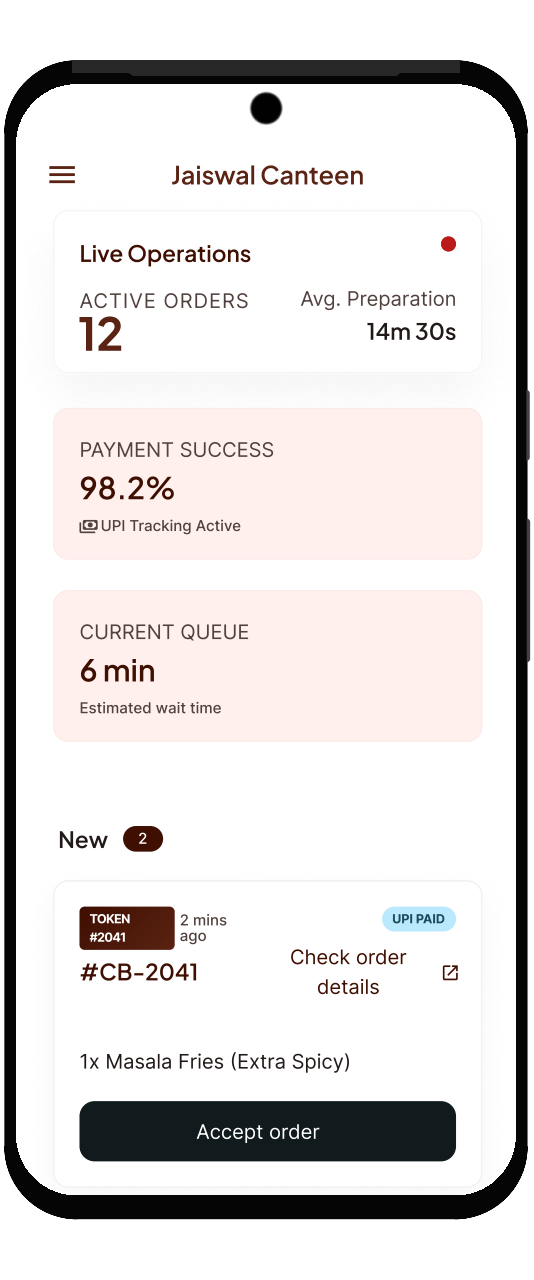

Jaiswal Canteen
Preparing Order
Cheese Grill Sandwich • 12:40 PM
10
mins left
Placed
Kitchen
Ready
Pickup
Order Summary
Cheese Grill Sandwich
1x • Extra Cheese
Rs.75

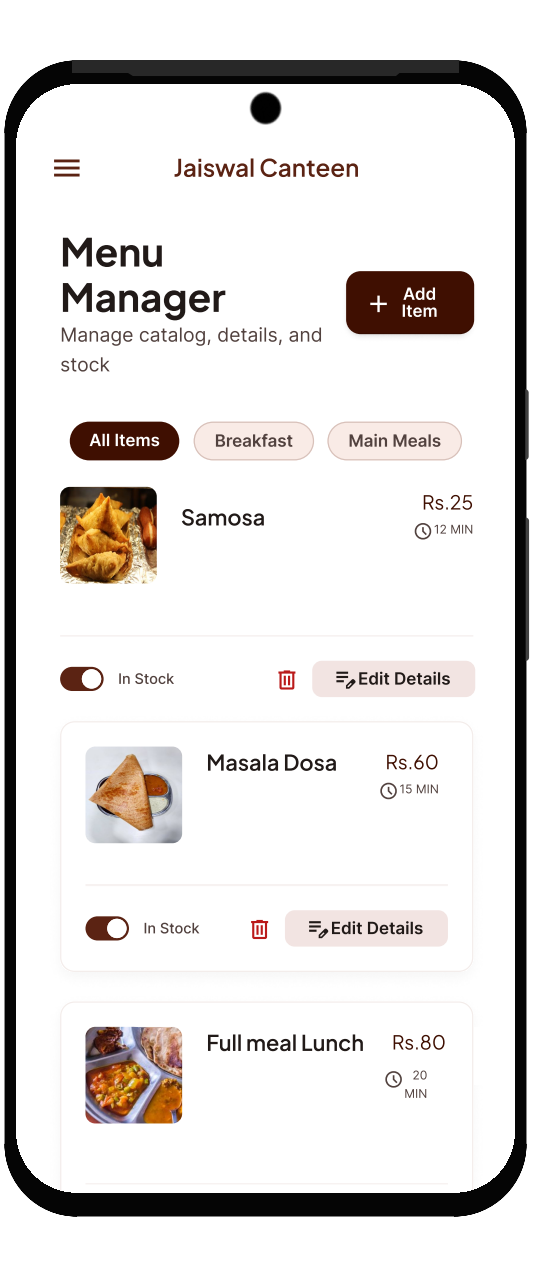

Jaiswal Canteen
Available Balance
250 Points
Redeem Rewards
Full Lunch Meal
5% Discount
Samosa
2% Discount
Noodles
Discount
Rewards

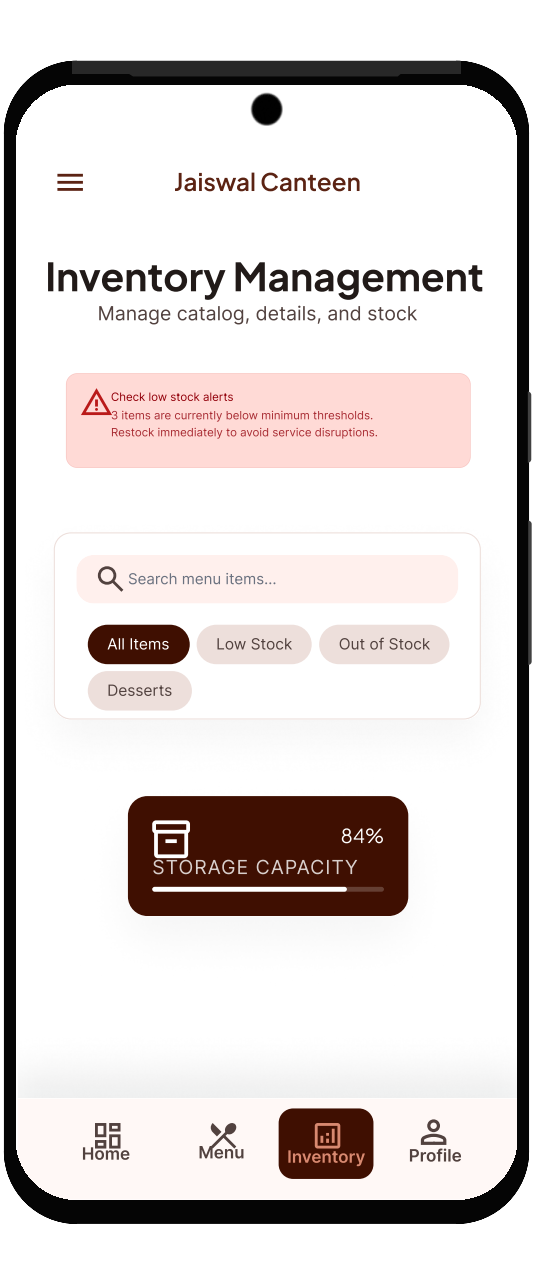

Jaiswal Canteen
Campus Customer
Activity
Recent Order
RECEIVED
Masala Dosa
April 12, 12:45 PM
Reorder
Details
Saved Items
Account
Personal Information
Address Book
Preferences
Notification Settings


Jaiswal Canteen
All
Breakfast
Meals
Snacks
Drink
CHEF'S SPECIAL
French Fries
Get 20% off today
₹90
Popular Dishes
View All
4.8
Paneer Wrap
Spiced paneer cubes wrapped in a soft whole...
Rs.80
4.5
Burger
Our signature veggie patty with house secret sauce.
Rs.60
4.9


Jaiswal Canteen
Hello, Admin
Here's what's happening at Jaiswal Canteen today.
Live Status:
Open
+12%
Today's Revenue
Rs. 12,450.00
82 Today
Total Orders
142
Active Orders
14
Live Orders (14)
View All
#42
Lunch Meal
Ordered 4 mins ago • Takeaway
Ready
#43
Burger, Fries
Ordered 8 mins ago •
Ready


Jaiswal Canteen
Live Operations
ACTIVE ORDERS
12
Avg. Preparation
14m 30s
PAYMENT SUCCESS
98.2%
UPI Tracking Active
CURRENT QUEUE
6 min
Estimated wait time
New
2
TOKEN #2041
2 mins ago
UPI PAID
#CB-2041
Check order details
1x Masala Fries (Extra Spicy)
Accept order


Jaiswal Canteen
Menu Manager
Manage catalog, details, and stock
Add Item
All Items
Breakfast
Main Meals
Samosa
Rs.25
12 MIN
In Stock
Edit Details
Masala Dosa
Rs.60
15 MIN
In Stock
Edit Details
Full meal Lunch
Rs.80
20 MIN


Jaiswal Canteen
Inventory Management
Manage catalog, details, and stock
Check low stock alerts
3 items are currently below minimum thresholds. Restock immediately to avoid service disruptions.
Search menu items...
All Items
Low Stock
Out of Stock
Desserts
84%
STORAGE CAPACITY
Home
Menu
Inventory
Profile